Sling Puck on the SO-101: Current Method and Simulation Results

CS 6341 project, Task 2 | Status on November 4, 2026 | Branch *slingpuck*, based on commit 1883db2, with changes that are not committed

1. Summary

The arm now wins most full-physics simulation matches at double speed. Greedy sling wins 18 of 30 matches at ×2 script speed and 6 of 30 at ×1. These results are from simulation only. The arm has not played a real match, and many values are placeholders.

Question	Current answer
What is the method?	Scripted arm skills, a shot model that chooses each shot, and a written rule for the game decision. PPO stays only as a challenger.
Can the arm shoot with a real friction grip?	Yes, in sim: the best shot passes the gate in 92 of 100 tries.
What limits speed?	×2 gives 96 of 100 shots and a 2.37 s cycle. Above ×2, the grip slips during the fast pull.
What does the touching start row cost?	The arm must push pucks out, and shots must cross the row. Clearing 5 pucks takes 16.6 s at ×2 speed.
Is the shot model correct?	Partly. It avoids the weak 1.5 cm pull now, but it is too confident on most other shots.
What is not known?	The real arm speed, real grip values, real scoring rule, and real human timing. 11–16 of each 30 matches time out, and the cause is not known yet.

No PPO training or other policy training ran for these results. The only fit was the small supervised shot model.

2. Task and goal

Two players shoot pucks through a gate in a center divider with elastic bands. The SO-101 must shoot and defend with one arm. Its sensors are one overhead webcam and the servo encoders, with no force sensing. We assume that the first side with no pucks wins, so the match is a **race of rates**:

net clearing rate = (shots per minute × P(through)) − (pucks received per minute)

Value	Used in sim	Source
Puck diameter, gate width	2.54 cm, 3.4 cm (launch limit ±4.3 mm)	Given by the team
Start row	Pucks touch: 5 per side, centers 2.54 cm apart	Given by the team
Row position	About 4.8 cm in front of the band	Placeholder: it varies in the real game
Scoring	First empty side wins; 60 s limit	Placeholder
Human	Reload 0.6 ± 0.2 s, place 0.2 s, pull 0.2 s, hold 0.05–0.3 s; half of the shots aim at the gate	Placeholder
Grip	Jaw friction 0.5, grip torque 1.47 Nm	Placeholder
Band, puck friction	Values in configs/physics_params_v0.yaml	Placeholder until real shots (S4)

3. Current method

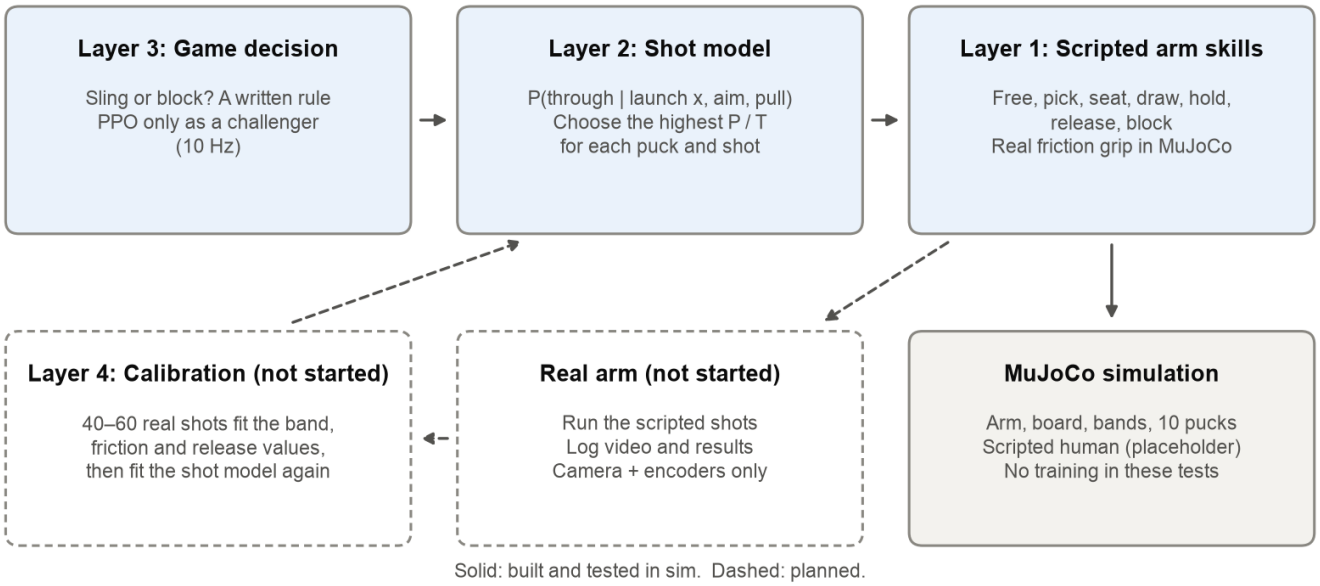


Figure 1. The four layers of the method. Layers 1–3 run in simulation. Layer 4 and the real arm are planned.

Layer 1: scripted arm skills

The arm uses inverse kinematics, with no neural network. Learned skills come later only if a script fails on the real arm.

Skill	What the arm does
Free	Pushes a puck out of the touching row with the two jaw tips. For a middle puck, it then opens its jaws in the hole to move the neighbors apart.
Pick	Rises, travels, waits until it arrives, descends with open jaws, closes, and checks that both jaws touch.
Seat, draw	Drags the puck to the band at launch x, then pulls it back by the chosen aim and pull.
Hold, release	Holds against the band force, then opens the jaws so that the band launches the puck.
Block	Puts the gripper in front of the gate to stop a shot.

Layer 2: shot model

The shot model predicts the chance that a shot passes the gate, from its launch x, aim and pull. It is an ensemble of 5 small neural networks, trained on 8000 simulated shots. Its held-out accuracy is 96.7%. It runs very fast, so the arm can compare 1470 candidate shots for each puck in each decision. The arm chooses:

the puck and shot with the highest P(through) / T, where T is the estimated time to the release.

T includes the pick, carry, seat, draw, and 0.8 s for a puck that must first be pushed out. The training shots include the measured jaw effect: exit speed × 0.875, exit x +1.1 mm, and exit angle +0.015 rad, plus noise.

Layer 3: game decision

A written rule decides when to sling or block. PPO may replace it only if PPO wins more matches. We compare these rules:

- **Greedy sling:** always shoots and never returns to the gate.
- **Greedy rule:** compares P/T for a shot with the threats that a block can save in the same time.
- **Reload rule:** shoots while the human reloads, and otherwise guards the gate.
- **Tell rule:** as the reload rule, and also shoots when the hand position shows a likely miss.

Layer 4: calibration (not started)

Log 40–60 real shots with a webcam at 120 fps or more. Fit the band, friction and release values, then fit the shot model again. Real grasp and release failures go back to the skills.

What changed from the midterm report

Part	Midterm report	Current method
Main question	When is it safe to leave the gate?	Which action clears the most pucks per second?
Game decision	PPO selector	Written rule; PPO only as a challenger
Shot	Fixed times and 95% success	Shot model with launch x, aim and pull
Grip in MuJoCo	Stiff constraint	Contact friction under a torque limit
Start row	Spaced pucks	Touching pucks, with a push-out skill

4. How we tested in simulation

- **Skill tests:** single shots in MuJoCo with a 2 mm camera error on the puck position. Each test uses 50–300 shots.
- **Row tests:** the arm clears all 5 pucks from the touching row, 45 runs per setting.
- **Event model:** 400 fast matches per point with placeholder arm times.
- **Full-physics matches:** the skills against the scripted human in MuJoCo, 30 matches per point.
- **Unit tests:** all 82 tests pass.

Three corrections affect how you read older results:

1 **Jaws and pucks did not collide before this work.** A collision filter was off, so the old MuJoCo match showed the block pose, not a physical block. The Task 2 MuJoCo description in the midterm report needs this correction.

2 **Friction now acts against the direction of motion.** The old per-axis friction removed small sideways speeds and hid angle errors at the gate. A straight 2.5 cm shot fell from 94 to about 67 of 100 after this fix.

3 **The start row now uses touching pucks.** The earlier result of 70.2% wins for greedy sling, and PPO's equal result, used spaced pucks and the old friction. They no longer apply.

5. Results

5.1 Full-physics matches

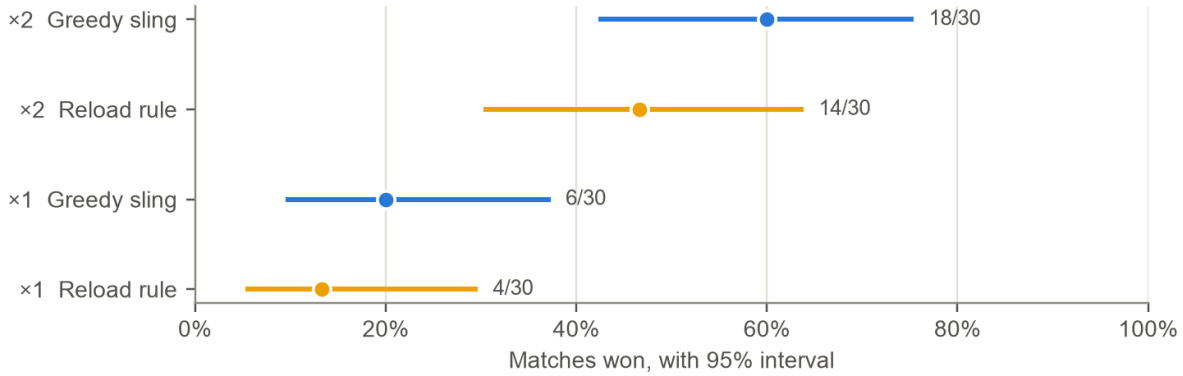


Figure 2. Wins in full-physics matches against the scripted human. Shot choice comes from the shot model. The intervals are wide with 30 matches.

Arm speed, rule	Wins	Losses	Time-outs	Mean match time	Robot / human shots
×2 Greedy sling	18	1	11	39.4 s	14.8 / 23.1
×2 Reload rule	14	3	13	47.5 s	13.7 / 28.1
×1 Greedy sling	6	9	15	46.7 s	13.8 / 27.4
×1 Reload rule	4	10	16	48.9 s	11.1 / 28.5

Speed has the largest effect. The two rules are not clearly different, because their intervals overlap.

5.2 Speed

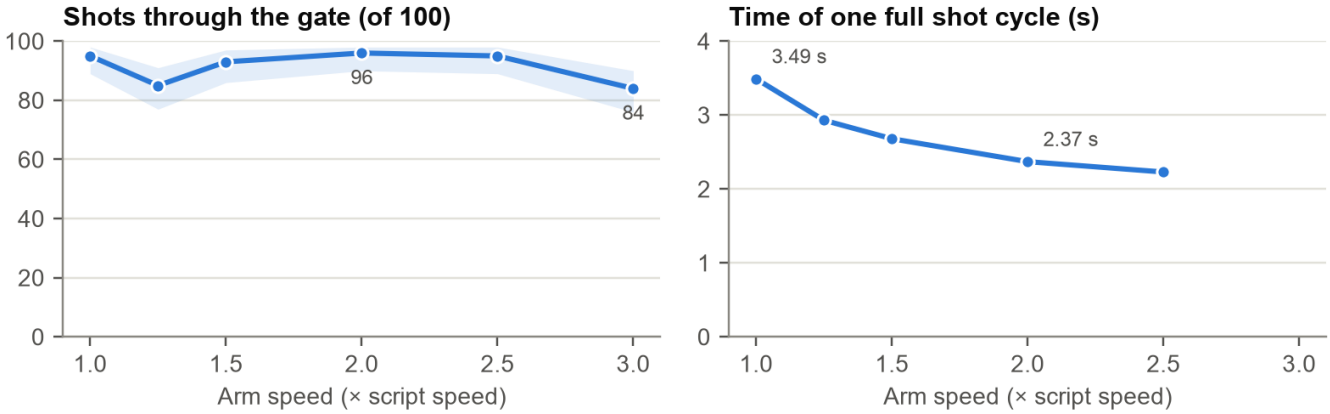
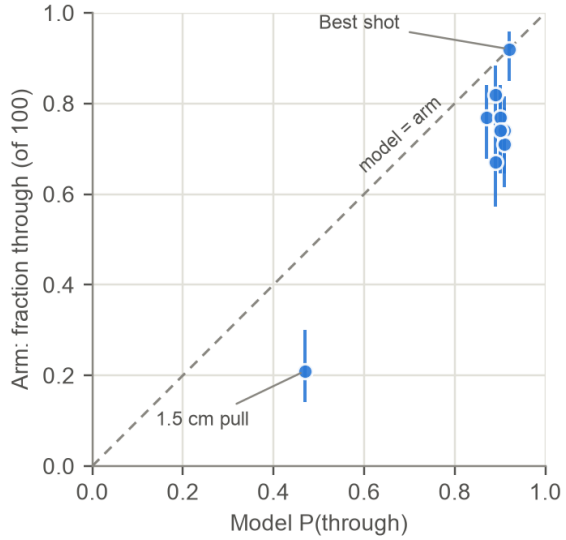


Figure 3. One shot at each speed (launch x +2.5 mm, aim −0.175 rad, pull 3 cm), 100 tries per speed. Cycle time covers shots with a good grip. At ×3, every grip slipped, so there is no cycle value. The shaded band is the 95% interval.

The arm first failed above ×1.25 because its low jaws hit the puck on the way from the gate. It now rises first, waits for arrival before each descent, and limits descent speed. At ×2.5, the jaw tips clear the board by only 0.6 mm and the grip slips in 65 of 100 shots, so ×2 is the practical sim limit. The dip at ×1.25 (85/100) is not explained.

5.3 Shot model against the arm



The refit fixed the largest error: the old model gave 0.93 for a 1.5 cm pull, and it now gives 0.47 (arm: 21/100). The model is still too confident on most shots:

Model P range	Mean P	Arm through
0.8–1.0	0.87	38/52
0.5–0.8	0.67	9/23
0.2–0.5	0.34	4/24
0–0.2	0.03	2/92

These bins use random shots with a good grasp. A straight 2.5 cm shot gets P = 0.89 but passes 67/100, and the best shot gets P = 0.92 and passes 92/100.

Figure 4. Model probability against the arm result for 9 shots, 100 tries each, with 95% intervals. Points below the dashed line show overconfidence. About 5 of each 100 tries are missed grasps, which the model does not include.

5.4 Clearing the touching row

Setting (45 runs each)	Time for 5 pucks	Tries	Passes per try
×2, no spread, opponent row in place	19.4 s	7.5	54%
×2, with spread, opponent row in place	16.6 s	6.6	69%
×2, no spread, opponent row moved aside	22.2 s	8.7	57%
×2, with spread, opponent row moved aside	19.6 s	7.4	73%
×1, with spread, opponent row in place	26.0 s	6.2	77%

Time ends when the last puck leaves the robot's half, and a puck can stop in the gate. One shot alone passes far more often than a shot from the row, because a launched puck must cross the row.

5.5 Other skill results at ×1

Test	Result
Pick	191 of 200 grasps. In the traced cases, misses came from a camera error of about 4 mm or more along the board.
Draw and release	Draw error 1.7 mm. Exit speed is 88% of the ideal band release.
Hold	1.6 mm drift in 2 s at a 3 cm pull, with no slip.
Block	Saved 15/15 at the gate center, 15/15 at 1 cm off, 11/15 at 2 cm off, 2/15 at 3 cm off, and 0/15 at 6 cm off (the control).
Joint calibration error	Best shot passes 93/100 at 0°, 90/100 at 0.25°, 90/100 at 0.5°, and 77/100 at 1°.
Jaw friction (of 50)	0.05: 13, 0.1: 13, 0.2: 37, 0.3: 46, 0.5: 45, 0.8: 46 passes.
Grip torque (of 50)	0.3 Nm: 3, 0.5 Nm: 45 (42 slips), 0.7 Nm: 45, 1.0 Nm: 46, 2.94 Nm: 35 passes. The drop at full torque is not explained.
Reach	All 169 test points on the agent half pass the top-down reach check. This check does not test arm collisions with the board.

5.6 Event model with placeholder arm times

Selector (400 matches, shot model)	Wins	95% interval	Losses	Time-outs
Greedy sling	16.5%	13.2–20.5%	30.2%	53.2%
Tell rule	6.5%	4.5–9.4%	55.8%	37.8%
Reload rule	1.0%	0.4–2.5%	76.8%	22.2%
Greedy rule	0%	0–1.0%	0%	100% (it always blocks)

This model uses placeholder arm times, so it is less realistic than Section 5.1. Its value is the timing sweep. The old fixed-time mode shows 29.5% wins for greedy sling, but it is not a fair comparison, because it ignores the push-out and assumes a 95% hit rate.

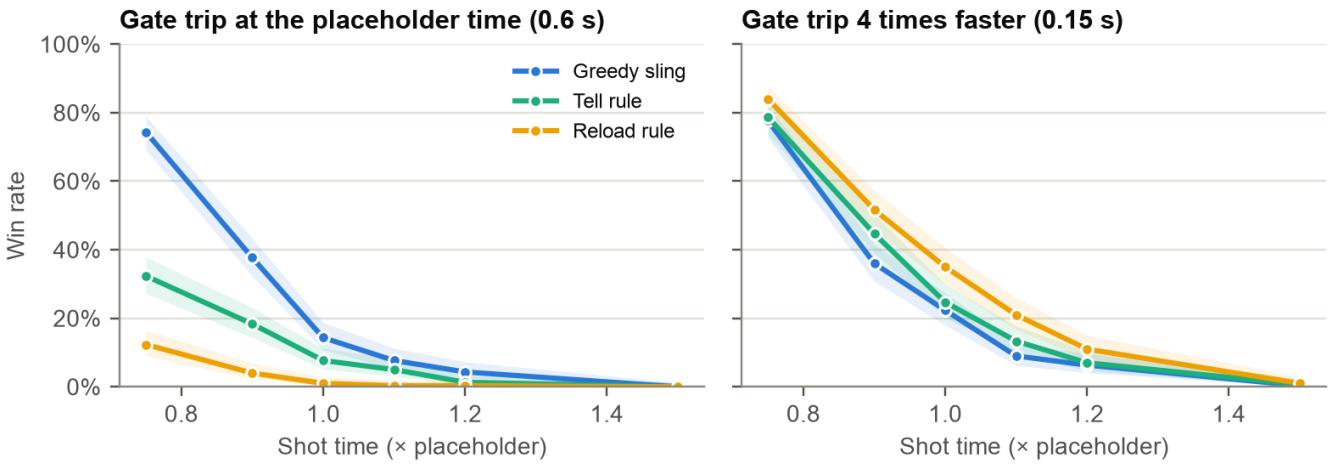


Figure 5. Event-model win rate against shot time, 300 matches per point, with 95% intervals. Half of the human shots aim at the gate.

A shot time 25% shorter raises greedy sling from 14% to 74% wins. A much faster gate trip makes the reload rule the best at most shot times, so blocking can matter if the real arm can return that fast. The current script takes about 1 s for that trip at ×1.

6. Limits of these results

- **Simulation only.** Real backlash, calibration drift, jaw wear and camera errors can lower every number.
- **×2 speed is not shown on the real arm.** The config stays at ×1 until the arm is tested.
- **The human is a script with placeholder times.** Real players can behave differently.
- **30 matches per point give wide intervals.** Use them to see large effects, not small ranks.
- **Time-outs are not diagnosed.** They can come from stuck or unreachable pucks, or from the rule itself.
- **The row position varies in the real game.** The tests use one position so far.
- **The placeholder jaw effect is built into the shot model.** Real shots must replace it.

7. Next steps

1. **Real-arm test (team):** Run the scripted shot at ×1, and raise the speed in steps. Record video and results.
2. **Scoring rule (team):** Confirm the real rule, and how a time-out counts.
3. **Time-outs (sim):** Find why they happen, then run more matches to rank the rules.
4. **Row positions (sim):** Test a range of row positions, because the distance varies.
5. **S4 calibration:** Log 40–60 real shots, fit the physics values, and fit the shot model again.
6. **Challenger:** Retrain PPO with the new skill times and rules, only after approval, and only if the written rules leave room to improve.
7. **S5 and S6:** Test against team-member profiles and humans, then deploy with webcam tracking.

Appendix: reproduce the results

Result	Command	Output folder
Skills	python -m src.slingpuck.eval.skill_eval	results/skills/s3_final/skills
Full-physics matches	python -m src.slingpuck.eval.mujoco_match_eval	results/match/s3_mujoco_matches
Event model	python -m src.slingpuck.eval.match_eval --selection greedy	results/match/s3_selection_refit
Timing sweep	python -m src.slingpuck.eval.timing_sweep	results/match/s3_sweep_refit
Shot model	python -m src.slingpuck.shot.model --shots 8000	cache/shot_model.npz

The speed, row, shot-check and grip tables came from one-off scripts. Their logs are in results/skills/s3_final/. The results folders are gitignored, and the code changes are not committed.